THE OHIO STATE UNIVERSITY
DEPARTMENT OF ENGINEERING EDUCATION

APPAM 2026 · BOSTON, MA

Teachers as *AI Policy Actors*: Focus Group Evidence from a K-12 Robotics Consortium

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01 BACKGROUND

Artificial intelligence tools are diffusing into K-12 STEM classrooms at a pace that outstrips corresponding federal guidance (Holmes et al., 2019; Akgun & Greenhow, 2022), while state agencies, school districts, and nonprofit consortia shape the terms of classroom AI use in fact. Teachers' needs, constraints, and attributions of accountability constitute an under-examined input to AI-in-education policy design.

02 THE E4USA+FIRST INITIATIVE

engineering 4 us all + FIRST

A federally funded, multi-state pre-college engineering initiative expanding access to robotics-based engineering learning. Flexible lessons serve as a stand-alone curriculum or supplemental content in high-school STEM courses.

Through partnerships between Engineering for US All (e4usa) and FIRST, the initiative positions teachers as co-creators and engages diverse urban, suburban, and rural contexts.

03 THEORETICAL LENS

STREET-LEVEL BUREAUCRACY (LIPSKY, 2010)

Teachers operating under policy ambiguity function as street-level bureaucrats whose discretionary judgments constitute a form of emergent policymaking.

04 RESEARCH QUESTION

As AI tools are introduced into pre-college instruction, what needs, constraints, and attributions of accountability do K–12 teachers articulate?

05 DATA

Three focus groups with K-12 teachers who participated in a federally funded multi-state robotics curriculum in the 2025–2026 academic year.

APRIL 2025 · FOCUS GROUP 1
AI-Tool Wishlist

JUNE 2025 · FOCUS GROUPS 2 & 3
After Hands-On Python-Curriculum Training

06 ANALYSIS

Transcripts were analyzed through deductive thematic analysis (Braun & Clarke, 2006), applying a three-tier coding framework that operationalizes Lipsky's (2010) street-level bureaucracy concept.

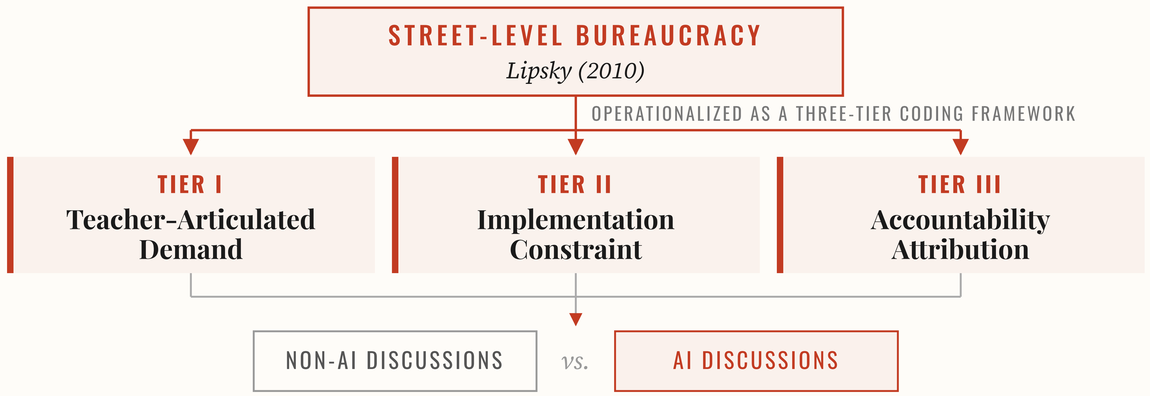


FIG 1. Three-tier coding framework operationalizing Lipsky's (2010) street-level bureaucracy concept.

07 FINDINGS · NON-AI VS. AI DISCUSSIONS

TIER I Demand

NON-AI DISCUSSIONS	AI DISCUSSIONS
Professional development was raised most frequently.	Demands concentrated on in-the-moment instructional assistance and curricular integration; professional development was mentioned far less often.

TIER II Constraint

NON-AI DISCUSSIONS	AI DISCUSSIONS
The most prominent constraint was time and capacity.	Policy and mandate ambiguity, together with tensions around pedagogical philosophy, became the most frequent constraints.

Equity-related constraints remained stable. Budget did not figure centrally in AI discussions.

TIER III Accountability

NON-AI DISCUSSIONS	AI DISCUSSIONS
Teachers rarely named industry.	Industry emerged as the most frequently mentioned actor, and school districts were named substantially more often.

Federal and state actors appeared only sporadically. Contextual differences were most pronounced in this tier.

Teachers frame AI primarily as support for their own technical work, such as code debugging, lesson design, and grading, rather than as a differentiation tool for addressing student heterogeneity.

08 IMPLICATIONS FOR STATE-LEVEL AI EDUCATION POLICY

01 · CLARIFY LEGITIMACY

Teachers' core perceived AI constraint is policy-legitimacy ambiguity. States and districts should prioritize clear, actionable classroom AI-use guidelines.

02 · DESIGN FOR EQUITY

Teachers' conceptions of AI have not yet connected to differentiated instruction, leaving room for equity-sensitive tool design for English learners and students with limited home connectivity.

03 · BUILD ACCOUNTABILITY

Industry has de facto become the most frequently identified actor for AI implementation, yet this role lacks an accountability framework. State and consortium regulators should develop arrangements that recognize industry–consortium collaboration as the dominant implementation mode.

YOUR FEEDBACK

Leave your thoughts here.

Akgun, S., & Greenhow, C. (2022). Artificial intelligence in education: Addressing ethical challenges in K-12 settings. *AI and Ethics, 2*(3), 431–440.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101.

Holmes, W., Bialik, M., & Fadel, C. (2019). *Artificial intelligence in education: Promises and implications for teaching and learning.* Center for Curriculum Redesign.

Lipsky, M. (2010). *Street-level bureaucracy: Dilemmas of the individual in public services* (30th anniversary expanded ed.). Russell Sage Foundation.

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